Hangulo: Demonstrating Thoughtful AI Support for Implementing Designers' Raw Ideas through Gradual Augmentation in Korean Lettering

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Fig. 1. Design scenario with **Hangulo**: A designer gradually augments raw ideas from rough sketches and a single letter into a complete Korean lettering set by leveraging AI features within their implementation workflow.

The design process is commonly understood as an iterative cycle of ideation, implementation, and evaluation. While prior research in AI-supported design has extensively explored AI assistance during ideation and evaluation, support for the implementation stage, where designers refine and develop their ideas into meaningful outcomes, remains underexplored. Recent MCP-based tool-calling agents enable task delegation through textual prompts but burden designers with deciding when and how to invoke the agent, and tend to automate implementation in a single step, overlooking the ongoing reflection during this stage. In this work, we present Hangulo, an AI-supported Korean lettering tool, to demonstrate that effective AI support for implementation need not require designers to explicitly control every action nor automate the entire process at once. Instead, such support can be designed to gradually augment designers' evolving inspirations within their existing workflows, preserving opportunities for micro-level reflection throughout implementation.

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CCS Concepts: • **Applied computing** → *Media arts*; • **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → **Intelligent agents**.

Additional Key Words and Phrases: lettering, typography, creative workflow, AI agents, implementation support, idea augmentation

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1 Introduction

Design processes typically unfold as iterative cycles of ideation, implementation, and evaluation. Prior research on AI-supported design has largely focused on the ideation [2, 6, 7, 17] and evaluation stages [3, 13], examining how AI can support activities such as reference exploration, recombination, and assessment while preserving designers' agency, autonomy, and controllability. In contrast, the implementation stage in between has received comparatively less attention. In practice, implementation is where designers translate raw ideas or low-fidelity prototypes into high-quality and meaningful outcomes. This stage demands substantial manual effort, time, and proficiency with design tools, as well as domain-specific creative know-how. As a result, a common pain point among designers is that even when they have strong ideas, they struggle with how to implement them effectively, which can discourage further exploration and stall iterative workflows.

Recent efforts to address implementation include embedding tool-calling AI agents within design tools, enabling designers to delegate tasks via textual prompts (e.g., "design a landing page based on this sketch") through MCP-based tool integration. While such approaches expand what agents can execute, they shift responsibility onto designers to determine when, how, and why agents should be invoked. Moreover, these approaches tend to treat implementation as automatable in a single step, overlooking that implementation involves ongoing reflection-in-action[11], where designers engage in micro-level ideation and evaluation. Rather than supporting designers' unfolding workflows, these agents require explicit task specification and management, revealing a gap in thoughtful AI integration that accounts for the incremental and reflective nature of implementation stage.

To explore this gap, we present a prototype, Hangulo, that demonstrates how AI can be integrated into the implementation process to support designers through gradual augmentation rather than one-shot automation. We ground our exploration in Korean lettering, a design practice that combines strong structural constraints with rich expressive possibilities. Rather than generating final designs or relying on textual prompts, Hangulo augments designers' raw expressions by leveraging workflow-native artifacts, such as edit history and sketches, as they emerge during implementation. The prototype consists of two main features: (1) *style spread,* in which the agent analyzes a stylistic change applied to a single phoneme letter and propagates it across other letters while adapting new letter shapes with consistency, and (2) *sketch apply,* in which the agent interprets designers' rough sketches and suggests multiple vector-level adaptations for a phoneme letter. Through this demonstration, we suggest that effective AI support for implementation should neither place the burden of orchestration on designers nor automate implementation in a single step. Instead, it should support the gradual augmentation of designers' evolving inspirations while preserving opportunities for micro-level reflection within existing implementation workflows.

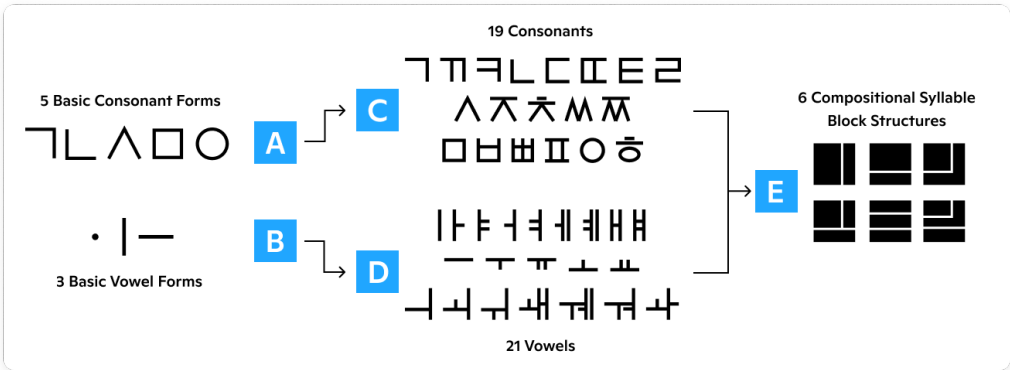

Fig. 2. Overview of the compositional structure of Hangul. Five basic consonant forms and three basic vowel forms act as visual primitives. These primitives expand into 21 consonants and 19 vowels. The resulting consonant and vowel elements are then arranged within six compositional layouts to form complete Hangul syllable blocks.

2 Background

2.1 Lettering and Korean Lettering

2.1.1 Lettering and Expressive Typography. Letters serve as a crucial tool in visual communication [5], encompassing the reading of text, interpretation of its meaning, and connections between text and other visual elements. Typography maintains a balance between aesthetic appeal and functional purpose. Lettering, often called expressive typography, emphasizes visual expression while accepting a reduced functional role [10]. It is the process of designing letterforms for specific words or phrases, allowing for creative expression while maintaining legibility within a particular context or purpose, such as titles for shows, movies, or albums. This contrasts with conventional typography, which prioritizes readability through careful arrangement of font, size, color, and layout, or with font design, which creates letterforms for general usage rather than specific content or contexts [9].

2.1.2 Hangul Lettering. Hangul, the Korean writing system described in Figure 2, consists of 19 consonants and 21 vowels [8]. Each consonant and vowel derives from 5 basic consonant forms and 3 basic vowel forms. Unlike Arabic letters, which combine linearly to construct syllables, Hangul's consonants and vowels assemble into 6 types of compositional block structures, with each completed block functioning as a single syllable. Due to these unique spatial characteristics and shared geometric features, Hangul requires more complex visual rules. For example, the shape of the same consonant varies depending on the composition of different syllable blocks. These characteristics also create distinctive creative opportunities in lettering, enabling designers to manipulate or exploit repetitive geometric structures in expressive ways.

2.2 AI Tools for Letter Design

Current AI approaches for letter design tend to focus on specific forms of stylization, such as deforming typefaces to create image-like calligrams [21, 22], synthesizing aesthetic typographic layouts [18], applying decorative textures to letterforms [1, 15, 16], or blending images and typography to produce expressive visual compositions [4, 12, 19, 20]. While these methods demonstrate the expressive potential of AI-assisted typography, they primarily operate at the level of image synthesis or high-level outcome steering [14], where designers adjust or guide generated results rather

than engaging in the construction of letterforms as structured typographic entities governed by properties such as aperture, axis, beak, or crotch. In this work, we focus on AI support for lettering that engages with the construction and development of letterforms during the implementation process. Rather than generating final typographic images, our approach emphasizes augmenting designers' raw expressions as they iteratively refine letterforms, supporting incremental and detail-oriented exploration within ongoing lettering workflows.

3 System Design

3.1 Design Process

To understand common practices and challenges in Hangul lettering, we reviewed online tutorials and classes, conducted a survey with practitioners (N=8), and participated in online lettering communities. The identified practices and challenges were validated through in-depth interview with a professional graphic designer who self-reported advanced Korean lettering proficiency (4/5).

Designers typically begin by sketching rough lettering ideas on grid paper or using digital drawing tools. When a promising sketch emerges, they move to vectorization using tools such as Illustrator, then iteratively refine the lettering through variations and adjustments. This process is non-linear: designers may return to sketching or proceed to vectorization even when ideas remain underdeveloped. We identified two recurring pain points. First, **designers struggle to extend an initial idea to unseen phoneme letters while balancing consistency and creative variation**. For example, when a designer develops a stylistic idea for the character 'ㅅ', adapting it to a related character like 'ㅈ', which shares a triangular basic form, is straightforward. However, adapting the same style to 'ㅁ', which derives from a different basic form, is considerably more challenging. Designers must reason about how to reinterpret strokes, proportions, and deformations across structurally different letters, often struggling to maintain visual coherence without becoming repetitive. Second, **modifying vectorized letterforms is cumbersome and error-prone**. Managing complex vector paths makes even minor adjustments time-consuming, discouraging exploration of small variations and interrupting the continuity of implementation. Building on these observations, we identified two design goals. First, optimize tedious aspects of vector manipulation. Second, preserve opportunities for micro-level reflection by gradually augmenting emerging ideas rather than replacing designers' decision-making.

3.2 Hangulo: An AI-Assisted Vector Editing Tool for Korean Lettering

Hangulo is a web-based vector editing tool designed to support Korean lettering by integrating AI assistance directly into designers' implementation workflows. It features six standard vector operations (Figure 3B) and two sketching operations (Figure 3B*). Once users input the target design word into the system, it constructs a layer for each syllable and each phoneme character. Users can import an existing base font or SVG file, or start from an empty canvas. The two AI features described below can be invoked by pressing the characters displayed at the bottom of the canvas (Figure 3E-F).

3.2.1 ***Style Spread.*** Style Spread enables designers to maintain stylistic consistency across Hangul letters by adaptively propagating style edits to characters that share similar structural forms. Rather than treating each letter as an isolated shape, the agent leverages Hangul's compositional structure to extend stylistic changes in a principled, controllable, and creative way. Once a designer modifies the style of a specific consonant or vowel, they can trigger this feature (Figure 3E). The interaction proceeds in four steps. First, the designer selects a stylistic change to propagate. Second, the agent presents its analysis and recommends a propagation range for other letters. During analysis, the

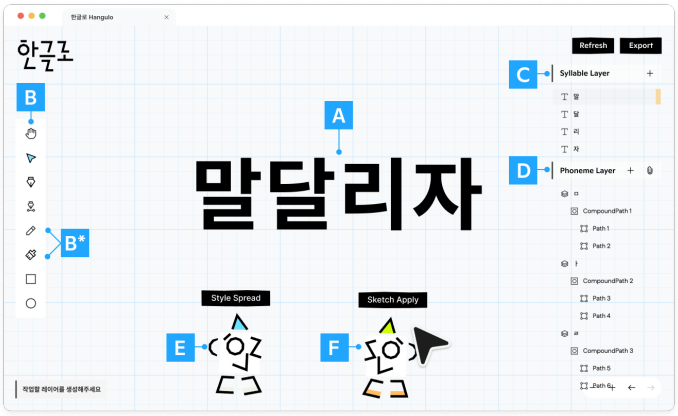

Fig. 3. Interface of **Hangulo**: (A) Vector canvas displaying the Korean word '말달리자', (B) Vector operation panel with six standard tools and (B*) two sketching tools, (C) Syllable layers of the target lettering word where the letter '말' is active, (D) Phoneme layers showing all vowels and consonants of the active syllable layer: 'ㅁ', 'ㅏ', 'ㄹ', (E) Style Spread agent, (F) Sketch Apply agent.

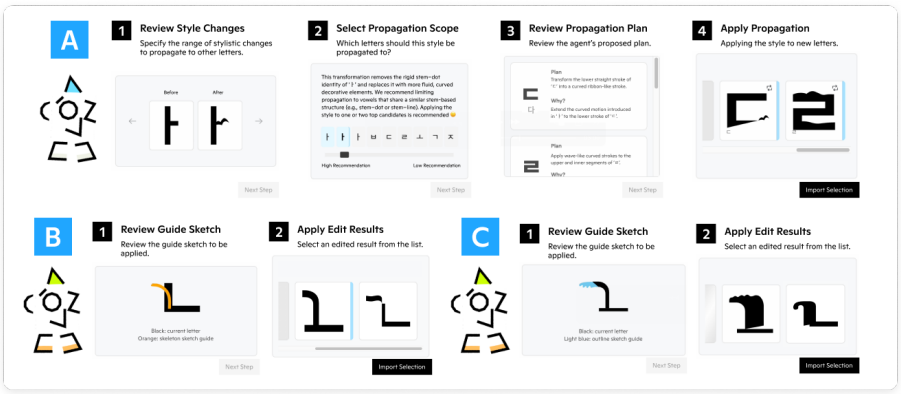

Fig. 4. Two core AI features of **Hangulo**: (A) Style Spread, (B) Sketch Apply with skeleton sketch using Marker tool, (C) Sketch Apply with outline sketch using Pencil tool.

agent identifies which structural features were altered (e.g., stroke curvature, thickness, or terminal shape) and detects other letters that share similar underlying forms, determining whether they are suitable targets for propagation. Once the designer confirms the propagation range, third, the agent generates a propagation plan for each phoneme letter that explains how it will be affected and why, allowing the designer to review the reasoning. Finally, the agent applies the planned stylistic transformations to the selected letters and presents the vector results for inspection. The designer can selectively import the generated vectors to the canvas or regenerate alternatives.

3.2.2 ***Sketch Apply.*** Sketch Apply (Figure 4B-C) enables designers to communicate raw stylistic intentions through informal sketches that guide the agent in generating deformed vectors. Designers can draw rough sketches directly over a phoneme letter on the canvas using two tools (Figure 3B*): Pencil and Marker.The Pencil tool is used when designers have an idea for the outline shape, while the Marker tool is used when they want to deform the stroke skeleton. For

instance, if a designer wants to add an expressive cap at the end of a stroke, they can sketch it with the Pencil tool. If they want to bend the stroke, they can indicate the skeleton shape with the Marker tool. After drawing sketches, designers can invoke the agent (Figure 3F) to apply them in various options. The agent interprets the intention behind the sketches and generates three different applications of the guide. It also generates alternative interpretations when the high-level intention can be understood in multiple ways.

3.3 Design Scenario

The designer Lee wants to create a New Year's greeting card for the Year of the Horse, one of the twelve animals in the East Asian zodiac. She decides to design the card with Korean lettering using the word '말달리자' (meaning "let's move forward with energy"). She first inputs the target word into Hangulo and loads a base font. She has a rough idea to make the design fluid and dynamic. She begins by sketching a curved vertical line on the vowel ' ㅏ' and visualizes this idea using *Sketch Apply* (Figure 1A). She selects the first option, which bends both ends of the vertical stroke, and decides to propagate it using *Style Spread* (Figure 1B). She applies the change only to vowels that share the same base form. While reviewing the current version, she realizes it would be more balanced with rounded corners, so she sketches a rounded corner on the consonant ' ㄷ ' and applies it with *Sketch Apply* (Figure 1C). She chooses the suggested version with oppositely rounded corners and propagates it to other consonants using *Style Spread* (Figure 1D). Looking at the complete lettering, she wants to simplify the design further, so she adjusts the consonant ' ㄹ ' and vowels by aligning them in the same direction. Finally, she polishes the details by adding subtle twists to some strokes (Figure 1E).

4 Implementation

Hangulo was built as a web-based application[1] using Paper.js[2]. To enhance the agent' s reasoning and performance, Hangulo incorporates heuristic representations of Hangul' s structural principles and shared geometric characteristics. Based on the eight basic forms (Figure 2) and their combinations, all consonants and vowels are categorized into one of nine form groups (e.g., square-group, circular-group, or vertical-line-group). The system further encodes morphological similarities using 18 form-level features (e.g., double-corner, diagonal-line, double-tick, or horizontal-line), enabling the agent to reason about shared geometric characteristics across Hangul consonants and vowels.

5 Conclusion

This work presented Hangulo, a design demonstration exploring how AI can support the implementation stage of creative practice by augmenting designers' evolving inspirations. Rather than generating final outcomes through textual instructions or requiring explicit task delegation, Hangulo integrates AI assistance into ongoing Korean lettering workflows by responding to intermediate artifacts such as edit histories and sketches. Importantly, Hangulo does not aim to automate implementation in a single step. Instead, the system supports gradual augmentation, allowing designers to expand small, localized inspirations into larger letter sets and evolving typographic concepts. These inspirations might include a stylistic change to a single letter or a rough sketch. By operating at this incremental scale, the system enables designers to continuously engage in micro-level ideation and evaluation throughout implementation.

This perspective opens several directions for future work. One agenda is to compare gradual implementation-level AI support with approaches that automate implementation in a single step, such as generating an entire lettering set from an initial sketch rather than supporting letter-by-letter development. Such comparisons could examine how different

[1]https://github.com/hye1ee/hangulo
[2]http://paperjs.org/

forms of assistance affect designers' sense of agency, authorship, and authenticity. More broadly, this work highlights that implementation itself involves ongoing micro-level ideation and evaluation. Supporting this stage therefore calls for incremental and staged forms of AI assistance rather than one-shot automation that collapses these reflective moments.

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